

Possible Near-Term Mission Architecture to Measure Lunar Polar Regolith Water Content

Stevie Nuss-Soeharto, Nathan K. Long and Graham E. Dorrington ¹

¹ *School of Engineering, RMIT University, Bundoora, VIC, 3083, Australia*

Summary: A possible follow-on to NASA's LCROSS mission is described. It is argued that, prior to lunar resource exploitation, it is necessary to better constrain the lunar regolith water-ice content and distribution in the southern pole region, to improve confidence that water can be found in sufficiently large quantities and to better inform the siting of future in-situ production facilities. A novel mission architecture solution is proposed comprising of three segments: a cis-lunar polar orbiter, a set of six impactors and a low lunar polar orbiting spacecraft. On successive orbits the cis-lunar orbiter releases one or more impactors such that each one strikes a different designated lunar polar target site with an impact velocity of ~2.5 km/s. Each impactor comprises of a ~0.3 m diameter hollow copper sphere fitted with a cold-gas micro-propulsion system to permit trajectory modification. Despite its poor structural properties, copper is selected to minimise measurement contamination. The ejector plume resulting from each impact is sampled by the orbiter, at a perilune altitude of ~50 km, using mass spectrometers (MASPEX and SUDA) and the impact site recorded using a thermal imager. In this manner, it is proposed that a robust measurement of the regolith-to-water ratio at multiple sites could be obtained. A preliminary mass and cost breakdown estimate of the proposed system is offered. It is concluded that the mission described might be achieved for ~\$500M USD (FY18), which is deemed lower cost than other possible architectures employing landers/hoppers/rovers. The overall mission risk is also characterised as being low, given that it builds directly on LCROSS experience and the use of existing scientific instrumentation.

Keywords: Moon, spacecraft, impactor, ice, mass spectrometer, plume, dust, vapour

Introduction

The existence of water-ice on the Moon is not only significant in terms of understanding the origins and evolution of the Moon and Solar System, but also offers the potential for in situ water production to be utilised for human exploration, in particular for life support systems and/or as a propellant for deep space missions [1-4].

As early as 1961, it was suggested that water could exist in the form of ice in the craters shadowed in permanent darkness around the lunar poles [5]. Evidence for this theory began to appear in the 1990s when, for example, the Lunar Prospector neutron data showed large quantities of hydrogen accumulated at the lunar poles [6]. Several subsequent missions were then launched to prove the presence of water around the lunar poles [7], [8], [9]. The Chandrayaan-1 lunar probe, launched by the Indian Space Research Organisation, used an M3 spectrometer to detect the signature of water ice in the Moon's northern and southern polar regions [7], [10]. Chandrayaan-1 validated the findings from the M3 spectrometer by launching a Moon Impact Probe, equipped with its own spectrometer (CHACE) which detected water vapour at high lunar latitudes during its descent towards the lunar surface. The Lunar Reconnaissance Orbiter (LRO), launched by NASA in 2009, supported these findings

by using radar to test the cross-section and circular polarization ratio (CPR) around the lunar poles [8]. Results from LRO showed high-CPR regions with signs indicative of the presence of water ice. While all the evidence collected by remote sensing was encouraging, in order to substantiate the results, a more direct observation of lunar water was desired. Launched in conjunction with the LRO, the Lunar Crater Observation and Sensing Satellite (LCROSS) was designed to provide an opportunity for direct observation of water on the Moon by generating a large debris plume from crashing the spent upper rocket stage of an Atlas V into the Cabeus crater, and then another by crashing itself into the lunar surface [9]. LCROSS ejected the Atlas V rocket stage into the Cabeus crater (located ~100km from the lunar south pole), then employed its thrusters to give it a four-minute separation between itself and the location above the point of impact so that it could fly through and analyse the resulting ejecta plume. The data collected from the plume was sent back to Earth before LCROSS intentionally deorbited, also colliding with the Cabeus crater, and generated a second ejecta plume. It was thought that the plumes created from the crashes would be sufficiently large to be observable on Earth, even using amateur telescopes, however, this proved to be incorrect. In the end, NASA was able to use the data sent back from LCROSS to arrive at a sub-surface water-to-regolith ratio estimate for the Cabeus crater of $5.6 \pm 2.9\%$.



Fig. 1: Artist's rendition of Centaur upper stage rocket approaching the moon with the Lunar Crater Observation and Sensing Satellite (LCROSS) [11].

While LCROSS provided an estimate for the water ice concentration levels present in one lunar crater, further missions are needed to determine whether, or not, the water-to-regolith ratio captured by LCROSS is representative of other water-containing lunar craters, before in situ production facilities can confidently set up.

This paper proposes a follow-up mission architecture for LCROSS, aimed at being relatively inexpensive (~\$500M USD) by using existing scientific instrumentation and technology, rendering it achievable in the near-future. The prospective mission is based on a novel, three-element architecture: a cis-lunar orbiter, a set of six impactors, and a low lunar polar orbiting spacecraft. The cis-lunar orbiter is needed to successively release six impactors at different craters close to the southern lunar pole, while the low lunar orbiter passes through the resulting ejecta plumes and analyses their composition using two mass spectrometers, MASPEX and SUDA, described later herein.

Proposed Mission Architecture

Concept of Operations

The proposed mission architecture is a novel approach which takes the goals of LCROSS to their next logical step. The architecture contains a three-component system: a cis-lunar polar orbiter, a set of six lunar impactors, and a low lunar polar orbiter. The three components will be launched together as payload on a launch system capable of achieving trans-lunar injection.

The orbital paths followed during this mission will resemble that of the LRO / LCROSS mission (Fig. 2). Once in cis-lunar space, the low lunar orbiter will be decoupled and manoeuvred into lunar orbit. The upper rocket stage, cis-lunar orbiter, and impactors will then assume a cis-lunar trajectory, orbiting around the Earth-Moon system until the appropriate angle is reached to successively release the impactors for collision with a series of lunar polar craters (Fig. 3), each generating an ejecta plume. The timing of the impactor release will match the orbital period of the low lunar orbiter such that it will pass through each debris plume on a separate orbit (Fig. 4). Table 1 identifies a possible selection of craters for this mission and gives their corresponding lunar coordinates and diameters. The choice of craters was made to minimise the longitudinal corrections required of the low lunar orbiter to be able to analyse consecutive plumes, as well as including Cabeus and Shackleton craters to validate the results from LCROSS and Chandrayaan-1 [7], [9].

LCROSS Orbit - Side View

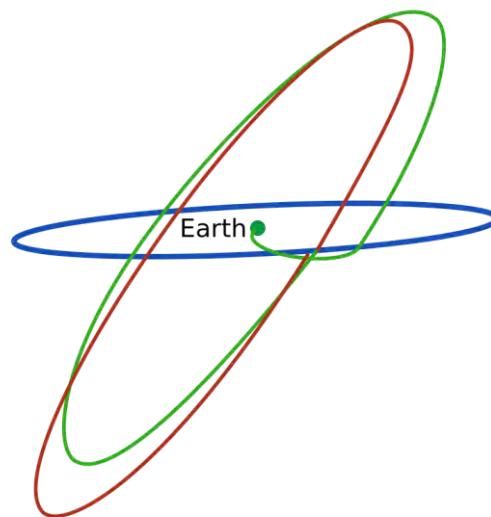


Fig. 2: LCROSS trajectory [12]: Moon's orbit, blue, LCROSS transfer orbit, green; final southern pole collision orbit, red.

The cis-lunar orbiter and upper stage potentially could then be deliberately crash landed onto the lunar surface to generate a further plume for analysis, as with LCROSS [9].

Table 1: Lunar craters [11].

Impact Crater	Lunar Coordinates	Diameter (km)
Cabeus	85.33°S, 42.13°W	100.58
Haworth	87.45°S, 5.17°W	51.42
Shoemaker	88.14°S, 35.91°E	51.82
Faustini	87.18°S, 84.31°E	42.48
De Gerlache	88.50°S, 87.10°W	32.40
Shackleton	89.67°S, 129.78°E	20.92

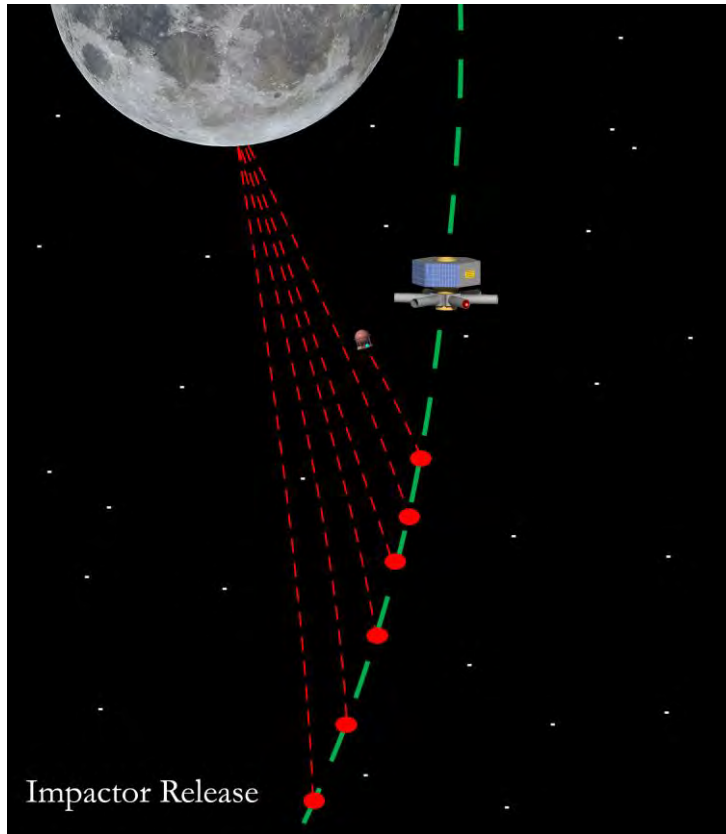


Fig. 3: Cis-lunar orbiter after releasing final impactor.

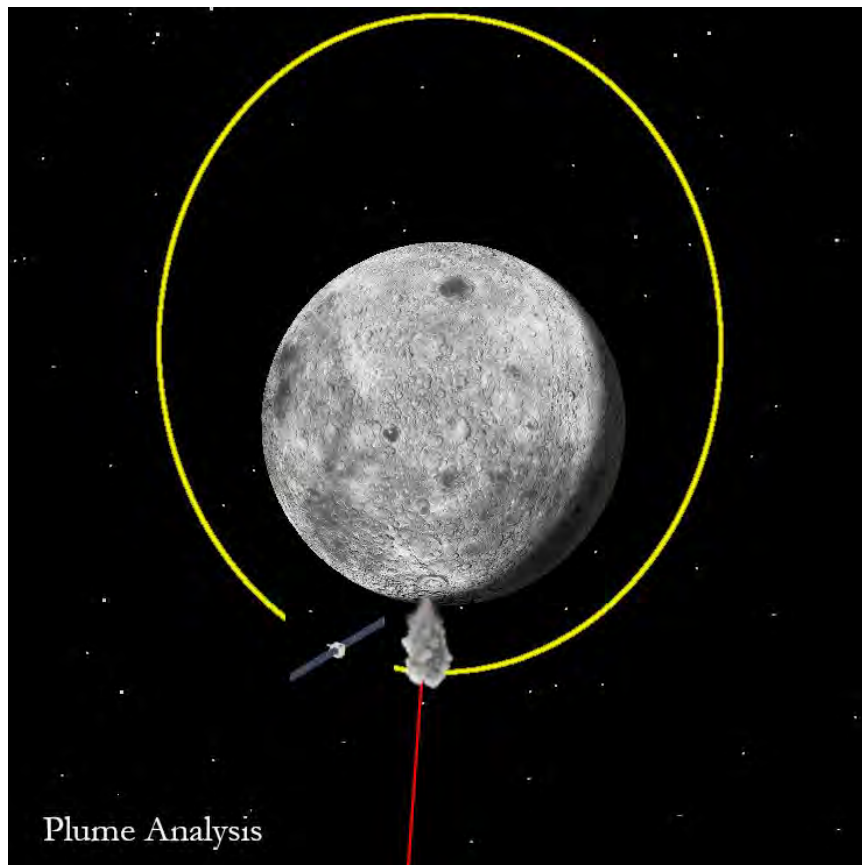


Fig. 4: Low-lunar orbiter about to analyse ejecta plume.

Cis-Lunar Orbiter

The cis-lunar orbiter (Fig. 5) is designed to store and launch six crater impactors (Fig. 6). On a similar orbital trajectory to the one used by LCROSS [9] the relative encounter velocity with the south lunar pole is of ~ 2.5 km/s. Up to six cis-lunar orbits are envisaged and one or more crater impactors are released on each orbit, a short time prior to lunar encounter. The concept design shown in Fig. 5 envisages six separate radially pointing launching tubes. Each tube houses one of the impactors and uses pressurised nitrogen to move a pneumatic ejection piston. Each piston imparts a delta-V of about 10 m/s on the impactor which then deviates away from the return orbit onto a new trajectory such that it hits its own individually prescribed lunar surface target with an impact velocity of ~ 2.5 km/s. A spacecraft bus attached to the launching tube assembly, provides the housing for the communications, power, and computing systems. At launch, the cis-lunar orbiter is mounted onto the upper stage of the (Falcon 9 or similar) launch vehicle and remains attached to this upper stage throughout the mission in order to utilise the stage's reaction control system.

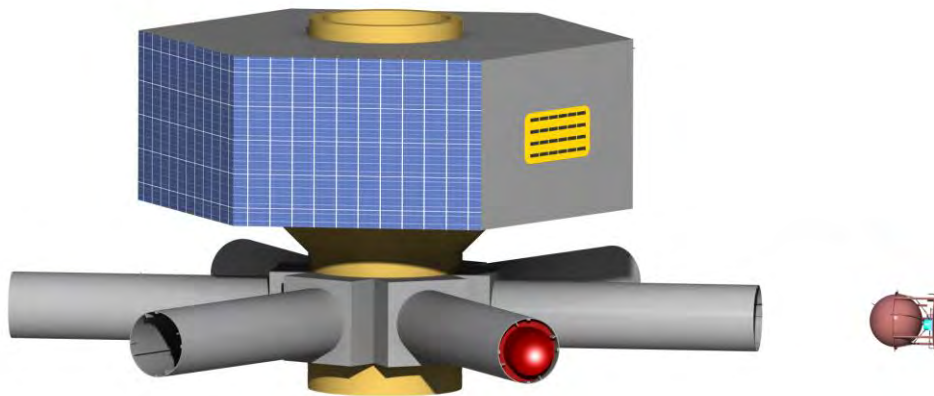


Fig. 5: Cis-lunar orbiter.

Impactor

Each impactor (Fig. 6) is comprised of a ~ 0.3 m diameter hollow copper sphere. This sphere constitutes most of the bulk mass needed for the impact (25 kg). It also provides a tank to contain xenon gas for a cold gas thruster system that is used to help steer the impactor to its selected surface target, see Table 1. Each impactor is also equipped with guidance and navigation instrumentation, UHF transceiver and an onboard power system. The mass of all these other systems is minimal (2 kg) and reduces the possibility of 'measurement contamination' of the lunar ejecta. Copper was selected as the main impactor material since it has a unique (easy to identify) mass spectrometer signature, despite having poor structural properties.



Fig. 6: Crater impactor.

Low-Lunar Orbiter

After departure from the cis-lunar orbiter and rocket stage, the low lunar orbiter will assume a lunar polar orbit with a perilune altitude of 50 km above the southern pole. Then, as each impactor ejector plume reaches an altitude above 50 km, the low lunar orbiter will pass through the plume and take samples using the mass spectrometers, while recording the impact site using a thermal imager. The impactors will collide into the selected craters at intervals devised to match the orbital periods of the low lunar orbiter, such that a separate plume analysis will be conducted on each orbit. This will help to minimise cross-contamination between craters and further validate or discredit the results from LCROSS. The low lunar orbiter (Fig. 7) will be comprised of a central bus which contains the power, communication, command and data handling, guidance and navigation, and instrumentation systems, as well as the propellant tank and thruster(s). Two foldable solar panels will be attached to the bus to provide the spacecraft with power throughout the mission, storing enough power in an onboard battery to keep the system online during orbital periods of darkness.

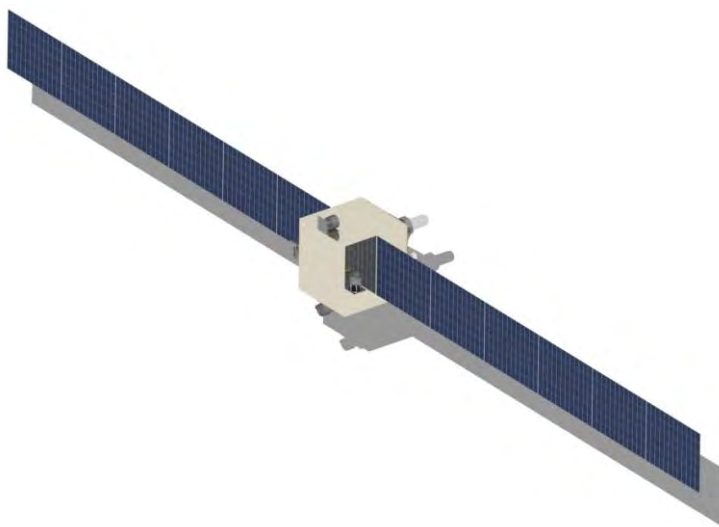


Fig. 7: Lunar Orbiter.

Measurement of Water-Regolith Ratio

The proposed mission aims build on the LCROSS mission by achieving more accurate results within a smaller tolerance. This will be executed by not only producing a larger sample size by creating six ejecta plumes but also accommodating more technologically advanced scientific instruments from NASA's Europa Clipper mission. During the planned flybys, the spacecraft will attempt to analyse the composition of the ejecta cloud particulates. Due to the location and composition of the blasted surface, it is assumed that the particulates will exist in both gaseous and solid dust form. For this reason, two NASA heritage mass spectrometers, SURface Dust Mass Analyzer (SUDA) [12] and MASSpectrometer for Planetary EXploration/Europa (MASPEX) [13], will be used to accurately perform the scientific analysis. Their performance specifications are highlighted in Table 2.

Table 2: Mass spectrometer performance specifications

	MASPEX	SUDA
Mass [kg]	8	4
Particle size [amu]	2-1,000	1-250
Mass resolution [m/dm]	7,000-24,000	150-200
Peak width definition	10% peak height	FWHM
Min particle speed [km/s]	-	4
Max particle speed [km/s]	-	30

The proposed imaging systems used on board will include a narrow and wide angle, visible spectrum cameras and a thermal, infrared spectrum camera. The benefit of acquiring products that are already deep into the developmental stage provides solid evidence of their functionality, therefore severely reducing risk of failure when compared to commercial off-the-shelf components. The primary purpose of the narrow angle camera will allow the generation high resolution images of the six selected craters before and after their respective impacts, as well as the ejecta plume conical spread geometries, while the wide-angle camera will be utilised to generate lower resolution colour images of the lunar surface map in regions surrounding the crater sites by taking advantage of its large image swath. The addition of the thermal imaging instrument will be to scan the crater sites for thermal anomalies before, during, and after impact, as well as generate complete thermal energy profiles, geometric validations, and geographic validations of the ejecta plumes.

The size of the impact crater generated by each impactor may be estimated using Holsapple's method for the "gravity regime" [14]. The crater volume, V , excavated by each impact is given by:

$$\frac{\rho V}{m} \approx K \left(\frac{ga}{U^2} \right)^{0.5} \left(\frac{\rho}{\rho_{\text{impactor}}} \right)^{-0.35} \quad (1)$$

where $m = 25$ kg is the impactor mass; $\rho = 1700$ kg/m³ is the regolith-ice mixture density; $g = 1.62$ N/kg is the lunar gravity; $a = 0.15$ m is the impactor radius; $U = 2$ km/s is the vertical velocity component of the impactor; ρ_{impactor} is the impactor effective density. This predicts a crater ejection mass of $\rho V \sim 3.6$ tonnes. The kinetic energy of the impactor is mainly released as heat, melting the copper impactor itself and vaporizing any volatiles trapped in the regolith to a predicted depth of ~ 1 m. The resulting ejecta comprises of regolith solid particulates as

well as the vaporized component. For LCROSS, Monte Carlo simulations were performed [14] to predict what particle mass fraction would reach altitudes in excess of 35 km. Based on the same proportionate scaling about 1%, or ~36 kg of material, would reach this altitude. This is considered more than sufficient to determine the water-to-regolith mass ratio.

Preliminary Mass and Cost Estimate

A preliminary mass and cost breakdown estimate of the proposed system is offered. It is concluded that the mission described might be achieved for ~\$500M USD (FY18), which is deemed lower cost than other possible architectures employing landers/hoppers/rovers.

Despite the intended use of heritage mission components and instruments, the system and architecture are complex in nature and will still require a large developmental phase. A work breakdown structure (WBS) of the total mission was formulated. This has been segregated into four major phases; planning, development, operations and launch, and post-mission, see Table 3.

Table 3: Mission cost breakdown.

Phase	Cost [\$M USD]
Planning	2.00
Development	391
Operations	53.0
Decommissioning	0.01
Post-mission	4.00
End mission	0.01
Mission Cost	449
Mission Cost + Insurance + Contingency	499

Based on known manufacturer mass budgets, a mass breakdown of the mission is given in Table 4. Similarly to the cost estimation, an additional contingency factor based on the product's TRL was added to the mass to account for any discrepancies that may occur during the development stage.

Table 4: Mass breakdown of mission.

Component	Mass [kg]
Cis-lunar spacecraft	DRY: 866
	WET: 874
Lunar orbit spacecraft	DRY: 137
	WET: 239
Total mission	1114

The mission requirement which would constrain the mass is the launch payload mass would be given by the total cost budget of the mission and the specific launch vehicle provider.

Comparison with Previous Missions

The LCROSS mission was estimated to have costed \$583M USD and LADEE at \$280 M USD, with launch masses of 621 kg and 382 kg, respectively [17, 18]. The mission proposed

is estimated to be of similar cost. The advance proposed over LCROSS will be the use of superior instruments that will analyse a larger impactor plumes to an improved mass tolerance, as well as the greater sample size of six South Pole craters. In reducing the % tolerance of certainty of the water-to-regolith ratio, the mission will increase the scientific value and knowledge of in-situ water located in each crater.

Conclusion

The proposed three-element mission architecture provides a viable solution to determining the water-to-regolith concentrations in the shadowed craters near the lunar poles that is achievable in the near-future. The novel architecture follows on from NASA's LCROSS mission by analysing multiple lunar crater impact ejecta using low-risk, existing instrumentation and technology proven on the LCROSS/LADEE and soon to fly EUROPA missions. This has the potential to speed up future lunar water extraction processes and to accelerate beyond LEO human space exploration.

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